

PROGRAM C2

C2. Explore the automobile dataset and visualize the i) Distribution of the two and four door cars with respect to the type of fuel they use ii) Distribution of cars of different body styles with respect to the type of fuel they use ii) Total number of each type of body style cars categorized by fuel type iii) Horsepower of each of the fuel type with reference to the type of drive wheel present in cars.

(<https://www.kaggle.com/datasets/toramky/automobile-dataset?resource=download>)

```
import pandas as pd
import matplotlib.pyplot as plt
data = pd.read_csv("Automobile_data.csv")
print(data.head(2))
```

```
   symboling  normalized-losses   make fuel-type aspiration num-of-doors \
0          3                ?  alfa-romero    gas         std         two
1          3                ?  alfa-romero    gas         std         two

   body-style drive-wheels engine-location  wheel-base  ...  engine-size  \
0  convertible         rwd         front      88.6  ...        130
1  convertible         rwd         front      88.6  ...        130

   fuel-system  bore  stroke  compression-ratio  horsepower  peak-rpm  city-mpg  \
0         mpfi  3.47    2.68                9.0         111     5000        21
1         mpfi  3.47    2.68                9.0         111     5000        21

   highway-mpg  price
0           27  13495
1           27  16500

[2 rows x 26 columns]
```

i) ***Distribution of the two and four door cars with respect to the type of fuel they use***

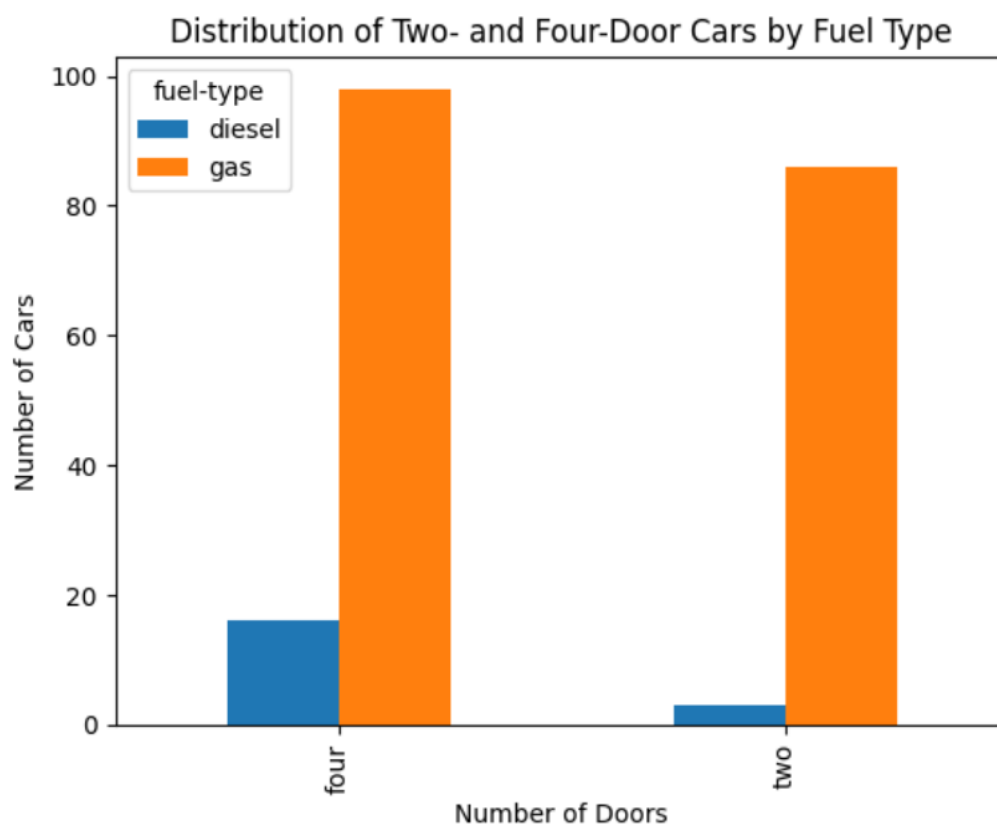
```
cars = data[['fuel-type', 'num-of-doors']].dropna()
cars = cars[cars['num-of-doors'] != '?']
print(cars.head())
```

```
   fuel-type  num-of-doors
0        gas           two
1        gas           two
2        gas           two
3        gas          four
4        gas          four
```

```
table = pd.crosstab(cars['num-of-doors'], cars['fuel-type'])  
print(table)
```

fuel-type	diesel	gas
num-of-doors		
four	16	98
two	3	86

```
table.plot(kind='bar')  
plt.title("Distribution of Two- and Four-Door Cars by Fuel Type")  
plt.xlabel("Number of Doors")  
plt.ylabel("Number of Cars")  
plt.show()
```



ii) ***Distribution of cars of different body styles with respect to the type of fuel they use ii)***

Total number of each type of body style cars categorized by fuel type

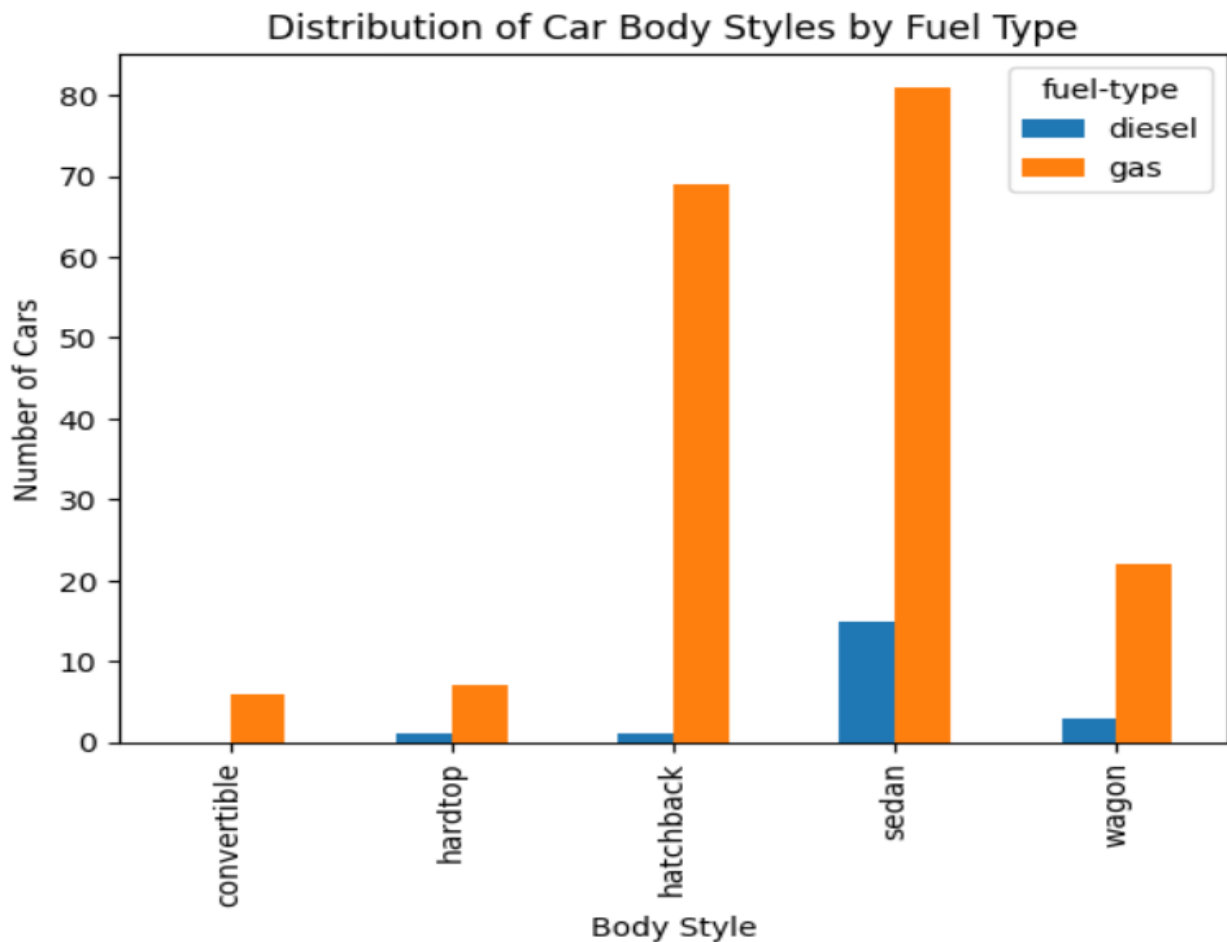
```
cars = data[['fuel-type', 'body-style']].dropna()  
print(cars.head())
```

	fuel-type	body-style
0	gas	convertible
1	gas	convertible
2	gas	hatchback
3	gas	sedan
4	gas	sedan

```
table = pd.crosstab(cars['body-style'], cars['fuel-type'])  
print(table)
```

fuel-type	diesel	gas
body-style		
convertible	0	6
hardtop	1	7
hatchback	1	69
sedan	15	81
wagon	3	22

```
table.plot(kind='bar')  
plt.title("Distribution of Car Body Styles by Fuel Type")  
plt.xlabel("Body Style")  
plt.ylabel("Number of Cars")  
plt.show()
```



iii) ***Horsepower of each of the fuel type with reference to the type of drive wheel present in cars.***

```
cars = data[['fuel-type', 'drive-wheels', 'horsepower']]
cars = cars[cars['horsepower'] != '?']
cars['horsepower'] = pd.to_numeric(cars['horsepower'])
print(cars.head())
```

	fuel-type	drive-wheels	horsepower
0	gas	rwd	111
1	gas	rwd	111
2	gas	rwd	154
3	gas	fwd	102
4	gas	4wd	115

```

table = pd.pivot_table(cars, values='horsepower',
index='fuel-type',
columns='drive-wheels',
aggfunc='mean')
print(table)
table.plot(kind='bar')
plt.ylabel("Average Horsepower")

```

drive-wheels	4wd	fwd	rwd
diesel	NaN	60.444444	104.090909
gas	95.333333	88.082569	138.707692

